LECTURE 9

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$n+1=m$

$\sum \frac{1}{n^2}$ conv. $\sum_{n=1}^{\infty} \frac{1}{(n+1)^2}$

Comparison test (Test 2): If $\sum a_n$ & $\sum b_n$ are +ve term series & $a_n \le b_n$

1) If $\sum b_n$ is cgt then $\sum a_n$ is cgt

2) If $\sum a_n$ is div then $\sum b_n$ is div.

Eg. 1) $\sum \frac{1}{\sqrt{5n^2+7n+11}}$

$5n^2+7n+11 \le 5n^2+7n^2+11n^2$

$\le 23n^2$

$\frac{1}{5n^2+7n+11} \ge \frac{1}{23n^2}$

$\frac{1}{\sqrt{5n^2+7n+11}} \ge \frac{1}{2\sqrt{n}} \cdot \frac{1}{n\sqrt{23}}$

a_n $\downarrow$ ∞

$b_n = \frac{1}{n\sqrt{23}}$ $\downarrow$ ∞

2) $\sum \frac{1}{n!}$

$2^{n-1} \le n!$

$1 = 1$

$2 = 2$

$4 \le 6$

$\frac{1}{2^{n-1}} \ge \frac{1}{n!}$

b_n a_n

$\sum b_n$ conv. (G.S.)

by ① $\sum a_n$ is cgt.

Limit Comparison test:

1) If $\sum a_n$ & $\sum b_n$ are +ve term series s.t. $\lim \frac{a_n}{b_n} \neq 0$ but finite then $\sum a_n$ & $\sum b_n$ converges & diverges together.

2) If $\lim \frac{a_n}{b_n} = 0$ Then if $\sum a_n$ div $\Rightarrow \sum b_n$ div & if $\sum b_n$ cgt $\Rightarrow \sum a_n$ cgt.

eg. 1) $\sum \frac{1}{\sqrt{5n^2+7n+11}}$ choose $b_n = \frac{1}{n}$

$$\lim \frac{a_n}{b_n} = \lim \frac{n}{\sqrt{5n^2+7n+11}} = \frac{n}{n\sqrt{5+\underbrace{\frac{7}{n}}_{\to 0}+\underbrace{\frac{11}{n^2}}_{\to 0}}}$$

$$= \frac{1}{\sqrt{5}} \neq 0 \text{ finite}$$

2) $\sum \frac{1}{n^2 e^n}$ choose $b_n = \frac{1}{n^2}$

$$\lim \frac{a_n}{b_n} = \lim \frac{1}{e^n} \to 0$$

Ratio & Root test:

If $\lim \frac{a_{n+1}}{a_n} = l$ $\quad l < 1$ cgt

$l > 1$ div

$l = 1$ test fails

(1) $\sum \frac{3^n}{n^3}$ $\qquad a_{n+1} = \frac{3^{n+1}}{(n+1)^3}$

$$\frac{a_{n+1}}{a_n} = \frac{3^{n+1}}{(n+1)^3} \cdot \frac{n^3}{3^n} = 3\left(\frac{n}{n+1}\right)^3$$

$\lim \frac{a_{n+1}}{a_n} = 3 > 1$ $\qquad \lim \frac{n}{n+1} = \frac{\cancel{n}}{\cancel{n}(1+\frac{1}{n})}$

$= 1$ $\qquad \frac{1}{n} \to 0$

$\sum a_n$ div.

2) $\sum \frac{n!}{n^n}$ $\qquad a_{n+1} = \frac{(n+1)!}{(n+1)^{n+1}}$

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \frac{n+1}{(n+1)^{n+1}} \cdot n^n$$

$$= \frac{n^n}{(n+1)^n}$$

$$= \left(\frac{n}{n+1}\right)^n$$

$$= \frac{1}{\left(1+\frac{1}{n}\right)^{n}} \to \frac{1}{e} < 1 \quad (\text{cgt})$$

$f(n)$, $g(n)$

$$\left[1 + f(n)\right]^{g(n)} = e^{f(n) \cdot g(n)}$$